

矩阵微分计算:

$$\textcircled{1} A'(t) = \frac{d(A(t))}{dt} = \left(\frac{d}{dt} a_{ij}(t) \right)_{m \times n}$$

②

$$\textcircled{2} \text{求导法则: i } (A(t) + B(t))' = A'(t) + B'(t)$$

$$\text{ii } (A(t) B(t))' = A'(t) B(t) + A(t) B'(t)$$

$$\text{iii } (\alpha A(t))' = \alpha \cdot A'(t)$$

$$\text{iv 若 } A'(t) \text{ 可微有 } (A'(t))' = -A'(t)A'(t)A'(t)$$

$$\text{v } (A(u))' = f'(t) \frac{d}{du} A(u)$$

$$\textcircled{3} (A^{(m)}(t))' = m A^{(m-1)}(t) \text{ 不一定成立, 因 } AA' \neq A'A$$

④ 初等函数求导: 设 A 与 t 无关

$$\text{i } (e^{tA})' = A e^{tA}$$

$$\text{ii } (\sin tA)' = A \cos(tA) = \cos tA A$$

$$\text{iii } (\cos tA)' = -A \sin tA = -\sin tA A$$

其他定义可近似可得

关于②-ii的证明

$$\text{记 } C(t) = (A(t) B(t)) = \begin{bmatrix} \alpha_1(t) \\ \vdots \\ \alpha_n(t) \end{bmatrix} [\beta_1(t) \cdots \beta_n(t)]$$

$$\text{则 } c_{ij}(t) = \langle \alpha_i(t), \beta_j(t) \rangle$$

$$\text{则 } c'_{ij}(t) = \sum_{p=1}^n [\alpha_{ip}(t) \beta_{pj}(t)]' \text{ 则可证}$$

$$= \alpha_{ip}(t)$$

证毕

关于②-iv的证明

$$A(t) \cdot A^{-1}(t) = I$$

$$\text{则 } A'(t) A^{-1}(t) + A(t) (A^{-1}(t))' = 0$$

$$\text{则 } \frac{dA^{-1}(t)}{dt} = -A^{-1}(t) A'(t) A^{-1}(t)$$